# Funkcje Analityczne (Analytical Functions)

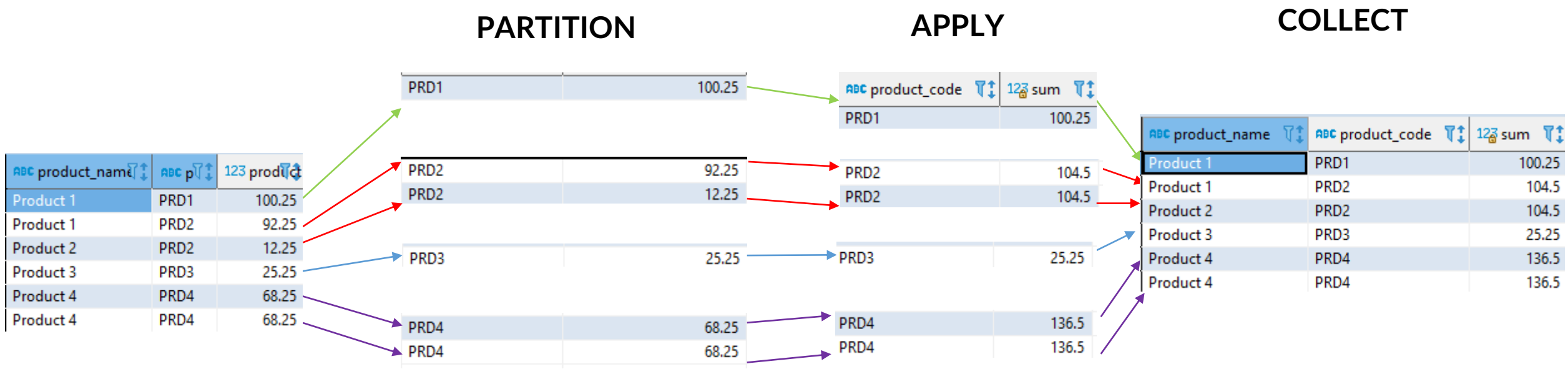
**Wartość liczona na podstawie grupy wartości zwraca dla każdego wiersza.**

- ROW_NUMBER
- LEAD / LAG
- FIRST / LAST VALUE
- RANK / DENSE RANK
- SUM / AVG / COUNT

```
SELECT *,
       FUNCTION() OVER (
                  PARTITION BY ...
                  ORDER BY ...)
 FROM ...
```

https://www.postgresql.org/docs/12/functions-window.html

# Schemat działania



# Schemat działania - rozszerzony

SOURCE → FILTER → PARTITION → SORT → FRAME → APPLY → COLLECT

```
function_name ([expression [, expression ... ]]) [ FILTER ( WHERE filter_clause ) ] OVER ( window_definition )
```

```
[ existing_window_name ]
[ PARTITION BY expression [, ...] ]
[ ORDER BY expression [ ASC | DESC | USING operator ] [ NULLS { FIRST | LAST } ] [, ...] ]
[ frame_clause ]
```

```
{ RANGE | ROWS | GROUPS } frame_start [ frame_exclusion ]
{ RANGE | ROWS | GROUPS } BETWEEN frame_start AND frame_end [ frame_exclusion ]
```

https://www.postgresql.org/docs/12/sql-expressions.html#SYNTAX-WINDOW-FUNCTIONS